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June 30, 1978
SUPERSEDING
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FEDERAL SPECIFICATION

CHARGERS, BATTERY, SEMICONDUCTOR RECTIFIER (FOR CHARGING INDUSTRIAL STORAGE BATTERIES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers self-contained single circuit battery charging equipment (rectifier type) for charging industrial storage batteries primarily used in materials handling equipment.

1.2 Classification.

1.2.1 Types and sizes. Unless otherwise specified, the chargers (see 6.2) shall be capable of charging the following battery types and sizes. If required, the ordering agency may specify battery voltages and capacities different from that listed below.

Charger type I - For nominal 36 volt batteries.
  Size A - Lead-acid 18 cells 450 to 850 AH.
Charger type II - For nominal 24 volt batteries.
  Size A - Lead-acid 12 cells 580 to 870 AH.
Charger type III - For nominal 12 volt batteries.
  Size A - Lead-acid 6 cells 360 to 580 AH.
  Size B - Lead-acid 6 cells 600 to 900 AH.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 6130

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc. (UL) Publication:

UL-1236 - Standard for Battery Chargers.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., Walt Whitmen Road, Melville, N.Y., 11749; 207 East Ohio Street, Chicago, Ill, 60613; or 1655 Scott Blvd., Santa Clara, Calif., 95050.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard product. Chargers covered by this specification shall be complete and ready for operation when delivered. Chargers shall be constructed so that they may be installed side by side requiring not less than 6 inches clearance on each side. When installed side by side, all operating controls, fuses, input and output connections shall be readily accessible from the front of the equipment. Structurally, the chargers shall be designed so that they may be readily lifted, removed or replaced in position by a fork lift truck or hoist, through the provision of suitable accessories. All major and minor assemblies, including accessories, shall be new and of standard design or model product of the manufacturer, except for such minor modifications from manufacturer's standard as may be required to conform to these specification requirements. The component parts of the unit need not be the products of the same manufacturer, however, the manufacturer shall be so established in the industry that prompt and continuing service and delivery of spare parts may be assured.

3.1.1 Material. Material shall be as specified herein. Material not definitely specified may be manufacturer's normal commercial product or the standard commercial component(s) used in the fabrication of the manufacturer's normal commercial product. "Normal commercial product" shall be interpreted to mean an end item produced by a manufacturer engaged in the commercial production of end item(s) covered by this specification.

3.1.1.1 Insulation. Insulating materials shall be capable of withstanding the temperature rise permitted for the class of insulation used, without evidence of breakdown or deterioration (see 4.3.3).

3.1.1.2 Electrical-resistance alloys. Electrical-resistance alloys shall be in accordance with the manufacturer's commercial practice.

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3.1.1.3 Structural steels. Structural steels shall be in accordance with the manufacturer's normal commercial practice.

3.1.1.4 Electrical sheet steels. Electrical sheet steels in laminated magnetic circuits shall be in accordance with the manufacturer's normal commercial practice.

3.1.2 Design. Design of chargers covered by this specification and all components therein, shall conform to accepted commercial design practices and shall insure safe, efficient and economical operation under normal service conditions. Chargers shall be single circuit full wave rectifiers with inherent voltage and current regulation. The charging cycle shall be automatically terminated at end of charge.

3.1.2.1 Input voltage. Unless otherwise specified (see 6.2), chargers shall be capable of operating on input voltage characteristics shown in table I within a tolerance of plus or minus 10 percent. The charger shall be capable of operating on input voltage of 208/230/460 by repositioning wiring taps and changing of the protective devices to the appropriate line current.

TABLE I

Type	Volts	Phase	Frequency
I, II, IV	208/230/460	3	47-63 Hz
III	208	Single	60 Hz

3.1.2.1.1 Input leads. Chargers shall be furnished with not less than 8 feet of UL approved input leads for connecting charger to the power supply. Leads shall be sized to conform with UL 1236.

3.1.3 Fire and casualty hazards.

3.1.3.1 Each contractor shall submit to the contracting agency proof that the chargers he proposes to supply under this specification conform to the requirements of UL 1236. The UL label, or listing shall be accepted as evidence that the chargers conform to this requirement.

3.1.3.2 In lieu of the label or listing, the contractor may submit independent proof, satisfactory to the contracting agency, that the charger conforms to the applicable requirements, including methods of tests of UL 1236.

3.2 Enclosures. Chargers shall be self-contained in a single enclosure with all necessary controls and instrumentation. Ventilating openings shall be so constructed that operation of the charger is not impaired. See 6.2 to specify special venting requirements (i.e. drip proof, or vents specifically located to conform with installation constraints).

3.3 Power transformer. The power transformer(s) shall be of the completely isolated multiple winding type. Transformers shall be designed so that the no-load exciting current will not exceed 10 percent of the maximum rated transformer current draw. When operated continuously in ambient temperature of 20 to 35°C at full rated load, the temperature rise shall not exceed for class A (55°C), class B (75°C), class F (115°C), and class H (150°C). The insulation and spacings shall be capable of withstanding without breakdown, the dielectric test in section 4.

3.4 Rectifier elements. Rectifier elements shall be of the silicon diode junction type. Temperature of diodes measured at the base shall conform to the requirements of U.L. 1236 when operating at rated load in an ambient of 20 to 35°C. Silicon diodes shall be Joint Electron Devices Engineering Council registered and shall have a manufacturer's peak inverse voltage rating at least 3 times the applied r.m.s. voltage at maximum charger output voltage. Heat sinks or radiators of aluminum may be used as conductors provided the plating on the diode is glavanically compatible. A heat conductive lubricant may be used for increased heat transfer and galvanic protection. Current carrying conductors for diode bases in main power circuits shall be copper or brass or plated aluminum.

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3.4.1 Cooling. Forced cooling of rectification elements shall not be used.

3.5 Operation. Chargers shall be capable of restoring 100 percent of the maximum ampere hours of the largets battery to be charged in 8 hours or less as defined in 3.5.1. Chargers shall be capable of operating continuously without exceeding allowable temperature rise specified herein, when operated in an ambient temperature of 20 to 35°C.

3.5.1 Lead-Acid batteries. Chargers shall have an inherent taper charging curve characteristic throughout the charge cycle. Two-step charging of batteries is not acceptable. Chargers shall be capable of returning a battery discharged its rated capacity to full charge in 8 hours or less without increasing electrolyte temperature more than 25°F. Electrolyte temperature during the entire charging cycle shall not exceed 43°C (110°F). A fully discharged battery is defined as one which has been discharged at the 6-hour rate to an end voltage of 1.70 volts per cell, or a battery with specific gravity readings of approximately 1.115 for flat pasted type, or 1.100 for multi-tubular type at 77°F (25°C) (or fully discharged gravity specified by the battery manufacturer). The starting rate shall be limited to not more than 25 amperes, or less than 16 amperes per 100 ampere hour rating of the battery and the final full charge voltage shall not exceed 2.55 volts per cell. The finish rate shall not exceed 5 amperes per 100 ampere hours of the rated battery capacity.

3.5.2 Efficiency. The conversion efficiency at rated output shall be not less than 60 percent for single phase and 70 percent for three phase units. Efficiency is based on the ratio of the d.c. power output to the a.c. power input measured at the charger terminal in such a manner to exclude d.c. cable or lead losses.

3.5.3 Voltage regulation. The inherent voltage regulation of the charger shall be such that the direct current output voltage shall not vary more than plus or minus 2 percent for an alternating current input voltage variation of plus or minus 10 percent over the entire charging range.

3.6 Auxiliary equipment. Each charger shall include the following auxiliary equipment: battery charging leads and charging connector(s); direct current ammeter; pilot light; electrically operated interval timer; and reverse current contactor, if the reverse current, with charger off, exceeds 1.0 percent of maximum rated output of charger.

3.6.1 Meters. Meters shall have an initial accuracy of plus or minus 5 percent.

3.6.2 Interval timer. The interval timer shall be the manually-reset type which incorporates initiating of the charging operation and automatic termination of the selected battery charging setting in its design. It shall be provided with a dial marked for selection of the daily battery charging setting and equalizing charge setting. The operation of the interval timer shall not affect any control of the inherent taper charging curve characteristic of the charger. Design of the timer shall provide for a maintenance-free operation during the life of the timer.

3.6.3 Battery charging leads and connector(s). Each charger shall be supplied with not less than 8-foot flexible output lead(s). Leads shall be sized to conform with UL-1236, with lugs on one end and specified charging connector(s) installed on the other end.

3.6.4 Pilot light. A neon type pilot light, visible from the front of the charger, shall be furnished to indicate when the charger is operating.

3.6.5 Protective features. Protective devices shall be provided in the primary circuit at the entrance to the cabinet to protect chargers from overload or burnout. The direct current output circuit shall be similarly protected. Means shall be provided to protect charging circuitry against damage from momentary voltage surges or spikes. The design of the charger shall provide a reverse polarity protection device to prevent damage in event the charging circuit is connected with the polarity reversed.

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3.7 Finish.

3.7.1 Treatment and painting. All exterior surfaces of the charger shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt, and rust, and shall be cleaned before painting. Exterior surfaces shall be painted per manufacturer's standard procedure to insure protection against rust and corrosion when operated or stored in damp atmospheric conditions. Primer and finish coats shall be applied in accordance with commercial practice to assure complete coverage and durability of finish. The finish coat, when dry, shall be a smooth even surface, free from runs, sags, or other defects. Surface of component and assemblies not normally painted with a finish coat shall be cleaned and prepared in accordance with good commercial practice.

3.8 Marking.

3.8.1 Components. All components of the chargers shall be identified with information normally employed on standard commercial products to facilitate replacement. The wiring for all circuits shall be color coded or clearly and durably numbered or lettered and all terminals marked for positive identification.

3.9 Technical manuals. When specified (see 6.2), one or more copies of the technical manual shall be furnished by the contractor with each charger at time of shipment. The manual shall be single-volume or supplementary and shall have contents as follows:

a. General description. The manual shall be applicable to each make and model of the charger furnished under the contract, including wiring diagrams, instructions for placing charger in service, including the individual battery tap connectors for ampere-hour starting rate settings, maintenance and repair, and parts list or parts breakdown. Manuals shall be attached to or packed with the charger when delivered.

3.10 Workmanship. Finished units shall be clean, well made, and free from any defects which may affect appearance or serviceability. All edges shall be smooth, rounded, and free from burrs or other irregularities. Unnecessary pores, pockets, and reservoirs, where oil, water, and dirt might collect, shall be eliminated.

3.10.1 Rivet connection. Rivet holes, in the sizes recommended by standard practice, shall be accurately punched or drilled, and shall have burrs removed. Rivet heads shall be full, neatly made, concentric with rivet holes, and in full contact with the surface of the member. Excessive upsetting of the rivet to fill holes will not be permitted.

3.10.2 Welds. Welded joints shall be sound, smooth, free from pits, holes, fissures, and rough projecting edges, and shall be thoroughly fused to the base metal.

3.10.3 Bolt connections. Bolt holes shall be accurately punched, drilled, reamed, or cast and adjacent edges shall have no burrs or fins to interfere with proper seating of bolthead or nuts.

3.10.4 Electrical connections. Electrical connections shall be clean and neatly accomplished. Soldered connections shall be thoroughly adherent to the base metal.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be made in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

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4.2.1 Visual examination. Each fully assembled unit shall be examined for completeness of assembly, presence and adequacy of components, technical manuals, if required, and quality of workmanship including wiring, soldered connections, welds, finishes, to assure that the unit is manufactured in accordance with specified requirements.

4.2.2 Testing of end item. Each production unit shall be tested in accordance with 4.3.3. Controls shall be checked to determine that they perform the functions for which they are intended. Failure to comply with these requirements shall be cause for rejection.

4.2.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.3 Test procedures. Tests to determine compliance with this specification shall be as described herein. Except as otherwise specified, all tests shall be conducted with chargers connected to applicable input supply voltage(s) specified in 3.1.2.1. No deviation to this requirement shall be permitted. Failure to meet these tests shall be cause for rejection of the unit(s) under test.

4.3.1 Endurance test. Chargers shall be subjected to 100 percent rated full load d.c. output for 6 hours. Temperature of transformer and rectification elements, output voltage and output current shall be recorded at 30 minute intervals. Temperature rise shall not exceed those permitted in 3.3 and 3.4. Load may be resistive in parallel with battery.

TABLE A. Battery charger inspections and tests

Inspection	Spec. (Req. para.)	Spec. (Insp. para.)	Inspection level	AQL (percent defective)
Visual	3.7, 3.8 3.10	4.2.1	S-2	2.5
Rated input and output	3.1.2.1	4.3	S-4	1.0
Voltage regulation	3.5.3	4.3.4	S-4	1.0
Efficiency	3.5.2	4.3.5	S-4	1.0
Function controls and timer	3.6; 3.6.1; 3.6.2	4.3.3.2	S-4	1.0
Dielectric test	3.3	4.3.3	S-4	1.0
Endurance	3.3; 3.4	4.3.1	S-4	1.0
Operation	3.5; 3.5.1	4.3.2	S-4	1.0
Other tests	3.6.5	4.3.6	S-4	1.0

4.3.2 Operation Test Procedure. Chargers shall be tested in accordance with the following test procedure to determine compliance with 3.5.1.

The chargers selected will be required to recharge a fully discharged battery of maximum ampere hour capacity specified within 8 hours (1 cycle each) as required by paragraph 3.5. This test is to be conducted at the manufacturer's plant at his expense and with his battery, or at a commercial laboratory acceptable to the Government at the manufacturer's expense. First test shall be conducted at 208 volt input, second test at 440 volt input, and third test at 208 volt input.

Step 1. Check technical manual furnished with charger to determine if charger is wired for correct ampere output of the battery being charged.

Step 2. Select a fully discharged battery or batteries totaling the specified maximum ampere hour rating of charger being tested. The battery or batteries selected shall have the ability to hold a 100 percent charge and the capability of delivering full rated capacity.

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Step 3. Do not start test until battery is fully discharged. The fully discharged battery shall have a specific gravity reading of approximately 1.115 for flat pasted plate and 1.100 for multi-tubular type (or fully discharged gravity specified by the battery manufacturer).

Step 4. Turn charger on and off three times with battery disconnected.

Step 5. With the fully discharged battery, the charger, and suitable instruments connected in the circuit, the charging circuit shall be closed.

Step 6. The AC input line current and voltage, DC battery current and voltage, electrolyte temperature and specific gravity readings shall be recorded during the charging cycle.

Step 7. Start test in morning and record charger ammeter readings at 30 minute intervals during the charging cycle to determine the charge taper curve. The battery should be fully charged (specific gravity approximately 1.255-1.260 at 77° F) within 8 hours. A slight specific gravity lag is normal, so approximately 1.255-1.260 might not be read for about 1/2 hour after the 8-hour charge is terminated. A specific gravity of 1.280 at 77°F (25°C) shall be observed after an equalizing charge is completed.

Step 8. Observe final battery voltage and specific gravity of electrolyte and compare with battery manufacturer's requirement for fully charged battery. The interval timer shall automatically terminate the charging cycle. Plot charging curve using information obtained from instruments.

Step 9. In order to determine the average ampere input for the 8-hour period, total the 16 ammeter readings recorded at 30 minute intervals (see step 7) and divide the result by 16. The average ampere input times 8 hours will show the total ampere hours replaced by the charger and should total at least 100 percent (or overcharge required by battery manufacturer) of the rated capacity (110% x 720 A.H. = 756 A.H.) without exceeding allowable battery temperature. Electrolyte temperatures during the entire charging cycle shall not exceed 43°C (110°F).

4.3.3 Dielectric test. The insulation and spacings of the silicon diode rectifier charger shall be subjected (for a period of one minute) to the application of a 60 cycle a.c. potential of 1000 volts plus twice the rated primary voltage between current-carrying parts of the primary circuit and (a) noncurrent carrying metal parts which may be grounded, and (b) current-carrying metal parts of an insulated secondary circuit. An insulated secondary circuit shall be subjected to a test voltage of 600 volts. There shall be no evidence of damage or insulation breakdown on a result of this test.

4.3.4 Voltage regulation. The charger shall be tested to determine compliance with 3.5.3. The regulation test shall be made with the maximum number of cells for which the charger is designed and rated. This test shall be conducted over an 8-hour period and may be conducted in conjunction with 4.3.2. Using a variac or other device, change rated input voltage each hour. The sequence shall be plus 10 percent rated voltage, and minus 10 percent rated voltage.

4.3.5 Efficiency. The charger shall be tested to determine compliance with 3.5.2.

4.3.6 Other tests. While charger is charging at maximum rated output, the charging connection shall be broken two or more times. Turn charger off and connect charging circuit with reverse polarity. Turn charger on momentarily and then off. The test unit other than activation of the protective devices cited in paragraph 3.6.5 shall show no damage as a result of these tests.

4.4 Code and standards. The supplier shall furnish evidence of compliance with applicable safety standards.

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letter should be issued to the supplier.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The packaging shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each charger shall be preserved and packaged in accordance with the normal commercial export practice. The complete package shall be designed to protect the charger against damage during shipment, handling and storage.

5.1.2 Commercial. Each charger shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. The packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Chargers packaged as specified in 5.1.1 shall be packed in export type containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification and the Uniform Freight Classification.

5.2.2 Commercial. Chargers packaged as specified in 5.1 shall be packed in fiberboard boxes that will assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling and storage. The shipping container shall be in compliance with National Motor Freight Classification and Uniform Freight Classification.

5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized to facilitate handling, in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Marking shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. Charging equipment covered by this specification is intended for use in recharging industrial storage batteries used in warehouse materials handling equipment such as forklift trucks, tractors, and handlift trucks.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, size charger required, and battery capacity, if different from 1.2.1.
(c) When a first article model is required (see 6.3).
(d) Alternate input voltage, when applicable (see 3.1.2.1).
(e) Special venting requirements (see 3.2).
(f) A number of technical manuals (see 3.9).
(g) Inspection responsibility, if other than specified (see 4.1).
(h) Level of packaging and packing required (see 5.1 and 5.2).
(i) Marking required (see 5.4).

6.3 First article model. When specified (see 6.2), a first article model shall be made ready for examination and testing in accordance with section 4 by the contracting officer or his authorized representative. The approved first article model may be presented for acceptance as the last unit for delivery under the contract. Evidence of prior testing of the charger type required, either the exact same model, or a similar model, falling within the size category required, shall be considered satisfactory evidence of compliance, unless the manufacturer had modified the design since it was previously tested.

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6.4 First article testing. The first article sample, if required, shall be tested for all provisions of this specification. Failure of the first article sample shall be cause for rejection of the unit under test. If a change is made in design, material or construction subsequent to approval of the first article sample, retest in accordance with 4.3 shall be required.

CUSTODIANS:

Army - EL
Navy - SA
Air Force - 82

Review Activities:

Army - EL, GL
Navy - SA, SH
Air Force - 82, 85

User Activities:

Navy - YD

Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

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